Net interest income and net yield excluding CIB's Markets businesses

In addition to reviewing net interest income and the net interest yield on a managed basis, management also reviews these metrics excluding CIB's Markets businesses, as shown below; these metrics, which exclude CIB's Markets businesses, are non-GAAP financial measures. Management reviews these metrics to assess the performance of the Firm's lending, investing (including asset-liability management) and deposit-raising activities. The resulting metrics that exclude CIB's Markets businesses are referred to as non-markets-related net interest income and net yield. CIB's Markets businesses are Fixed Income Markets and Equity Markets. Management believes that disclosure of non-markets-related net interest income and net yield provides investors and analysts with other measures by which to analyze the non-markets-related business trends of the Firm and provides a comparable measure to other financial institutions that are primarily focused on lending, investing and deposit-raising activities.

Year ended December 31, (in millions, except rates)	**2018**	2017	2016
Net interest income – managed basis[(a)(b)]	**$ 55,687**	$ 51,410	$ 47,292
Less: CIB Markets net interest income[(c)]	**3,087**	4,630	6,334
Net interest income excluding CIB Markets[(a)]	**$ 52,600**	$ 46,780	$ 40,958
Average interest-earning assets	**$2,229,188**	$2,180,592	$ 2,101,604
Less: Average CIB Markets interest-earning assets[(c)]	**609,635**	540,835	520,307
Average interest-earning assets excluding CIB Markets	**$1,619,553**	$1,639,757	$ 1,581,297
Net interest yield on average interest-earning assets – managed basis	2.50%	2.36%	2.25%
Net interest yield on average CIB Markets interest-earning assets[(c)]	0.51	0.86	1.22
Net interest yield on average interest-earning assets excluding CIB Markets	3.25%	2.85%	2.59%

(a) Interest includes the effect of related hedges. Taxable-equivalent amounts are used where applicable.
(b) For a reconciliation of net interest income on a reported and managed basis, refer to reconciliation from the Firm's reported U.S. GAAP results to managed basis on page 57.
(c) For further information on CIB's Markets businesses, refer to page 69.

Calculation of certain U.S. GAAP and non-GAAP financial measures

Certain U.S. GAAP and non-GAAP financial measures are calculated as follows:

Book value per share ("BVPS")
Common stockholders' equity at period-end / Common shares at period-end

Overhead ratio
Total noninterest expense / Total net revenue

Return on assets ("ROA")
Reported net income / Total average assets

Return on common equity ("ROE")
Net income* / Average common stockholders' equity

Return on tangible common equity ("ROTCE")
Net income* / Average tangible common equity

Tangible book value per share ("TBVPS")
Tangible common equity at period-end / Common shares at period-end

* Represents net income applicable to common equity

The Firm also reviews adjusted expense, which is noninterest expense excluding Firmwide legal expense and is therefore a non-GAAP financial measure. Additionally, certain credit metrics and ratios disclosed by the Firm exclude PCI loans, and are therefore non-GAAP measures. Management believes these measures help investors understand the effect of these items on reported results and provide an alternate presentation of the Firm's performance. For additional information on credit metrics and ratios excluding PCI loans, refer to Credit and Investment Risk Management on pages 102-123.